```
import random

# Training text - the Markov Chain will learn from this
text = """
The sun rose over the mountains casting golden light across the valley.
The birds began to sing as the morning mist slowly faded away.
She walked through the forest listening to the rustling of leaves.
Every tree told a story of centuries gone by.
The river flowed gently carrying with it the memories of the land.
The stars appeared one by one as the evening sky darkened.
He sat by the fire lost in thoughts of distant places.
The wind carried the scent of rain across the open fields.
"""

print("Text loaded!")
```

```
Text loaded!
```

```
# Build the Markov Chain
def build_markov_chain(text):
    words = text.split()
    chain = {}

    for i in range(len(words) - 1):
        current_word = words[i]
        next_word = words[i + 1]

        if current_word not in chain:
            chain[current_word] = []

        chain[current_word].append(next_word)

    return chain

chain = build_markov_chain(text)
print("Markov Chain built!")
print("Vocabulary size:", len(chain), "words")
```

```
Markov Chain built!
Vocabulary size: 65 words
```

```
import random

# Training text
text = """
The sun rose over the mountains casting golden light across the valley.
The birds began to sing as the morning mist slowly faded away.
She walked through the forest listening to the rustling of leaves.
Every tree told a story of centuries gone by.
The river flowed gently carrying with it the memories of the land.
The stars appeared one by one as the evening sky darkened.
He sat by the fire lost in thoughts of distant places.
The wind carried the scent of rain across the open fields.
"""

# Build chain with 2 words at a time
def build_chain_2(text):
    words = text.split()
    chain = {}
    for i in range(len(words) - 2):
        key = (words[i], words[i+1])
        next_word = words[i+2]
        if key not in chain:
            chain[key] = []
        chain[key].append(next_word)
    return chain

def generate_text_2(chain, start1, start2, length=20):
    result = [start1, start2]
    for _ in range(length):
        key = (result[-2], result[-1])
        if key in chain:
            next_word = random.choice(chain[key])
            result.append(next_word)
        else:
            break
    return " ".join(result)

chain2 = build_chain_2(text)
```

```
print("--- Output 1 ---")
print(generate_text_2(chain2, "The", "sun", length=20))

print("\n--- Output 2 ---")
print(generate_text_2(chain2, "She", "walked", length=20))

print("\n--- Output 3 ---")
print(generate_text_2(chain2, "The", "river", length=20))
```

```
--- Output 1 ---
The sun rose over the mountains casting golden light across the valley. The birds began to sing as the evening sky darkened.

--- Output 2 ---
She walked through the forest listening to the rustling of leaves. Every tree told a story of centuries gone by. The river

--- Output 3 ---
The river flowed gently carrying with it the memories of the land. The stars appeared one by one as the evening sky
```

```
def generate_text_complete(chain, start1, start2, length=30):
    result = [start1, start2]
    for _ in range(length):
        key = (result[-2], result[-1])
        if key in chain:
            next_word = random.choice(chain[key])
            result.append(next_word)
        else:
            break

    sentence = " ".join(result)

    # Only return if sentence ends with a proper punctuation
    if sentence.endswith(".") or sentence.endswith("!") or sentence.endswith("?"):
        return sentence
    else:
        # Add a period at the end to complete it
        return sentence + "."

print("--- Output 1 ---")
print(generate_text_complete(chain2, "The", "sun", length=30))

print("\n--- Output 2 ---")
print(generate_text_complete(chain2, "She", "walked", length=30))

print("\n--- Output 3 ---")
print(generate_text_complete(chain2, "The", "river", length=30))
```

```
--- Output 1 ---
The sun rose over the river as the birds sang their morning songs. She listened to the river flowing gently through the vall

--- Output 2 ---
She walked through the forest listening to the river as the birds sang their morning songs. She listened to the river flowir

--- Output 3 ---
The river flowed gently carrying with it the memories of the land. The stars appeared one by one as the stars began to sing
```

```
def generate_text_complete(chain, start1, start2, length=30, min_words=10):
    for attempt in range(10):  # try up to 10 times
        result = [start1, start2]
        for _ in range(length):
            key = (result[-2], result[-1])
            if key in chain:
                next_word = random.choice(chain[key])
                result.append(next_word)
            else:
                break

        # Only return if sentence is long enough
        if len(result) >= min_words:
            return " ".join(result) + "."

    return "Could not generate a complete sentence."

print("--- Output 1 ---")
print(generate_text_complete(chain2, "The", "sun", length=30, min_words=15))

print("\n--- Output 2 ---")
print(generate_text_complete(chain2, "She", "walked", length=30, min_words=15))

print("\n--- Output 3 ---")
print(generate_text_complete(chain2, "The", "river", length=30, min_words=15))
```

```
--- Output 1 ---
The sun rose over the river flowing gently through the valley below. The evening light faded slowly as the morning mist slow

--- Output 2 ---
She walked through the open fields carrying memories of distant lands. The morning mist slowly faded away. She walked throug

--- Output 3 ---
The river flowed gently carrying with it the memories of distant places. The wind carried the scent of rain across the valle
```